



Experimental investigation and thermodynamic modeling of the Ti–V–Zr system



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ABSTRACT

The isothermal sections at 1073 and 1273 K, and liquidus surface projection of the Ti–V–Zr system were investigated using X-ray powder diffraction (XRD), scanning electron microscopy (SEM) with energy dispersive spectrometer (EDS) and differential thermal analysis (DTA). Combined with experimental results in the present work and literature, the Ti–V–Zr system was optimized by the means of the CALPHAD (CALculation of PHase Diagram) method. The solution phases, liquid, hcp (Ti, Zr) and bcc (Ti, V, Zr), were modeled with the substitutional solution model, and the compound V_2Zr was treated as (Ti, V, Zr)₂(Ti, V, Zr) based on its thermodynamic model in the V–Zr system and the experimental solid solubility of Ti in the Ti–V–Zr system. A set of self-consistent thermodynamic parameters of the Ti–V–Zr system was obtained.

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1. Introduction

Hydrogen as a renewable substitute for fossil fuels has attracted much attentions in recent years. Among various types of hydrogen storage alloys, the Ti–V-based alloys with body centered cubic (bcc) structure have been identified as promising materials for hydrogen storage applications, based on their remarkable hydrogen-absorbing capacity [1–3]. However, these alloys suffer from certain disadvantages like difficulty in the activation for hydrogen absorption, slow kinetics, large hysteresis, low cyclic durability and high cost [4,5]. To resolve these problems, alloy elements substitutions have been attempted at the Ti site as well as the V site [6–8]. Zr substitution was reported to improve the hydrogen absorption properties of Ti–V-based alloys such as a decrease of the hysteresis and plateau pressure and an increase of the width of the plateau by improving the grain boundary network of Laves phase [9–11].

In order to identify optimal compositions, it was necessary to analyze the phase diagram, and subsequently to derive the thermodynamic data for all the phases in the system and their variation with temperature and composition. However, there was only limited phase equilibria information in the Ti–V–Zr ternary system [12,13]. In order to rectify this and to provide a firm basis for the thermodynamic assessment of this system, the key experimental work was carried out by XRD, SEM-EDS and DTA measurements in the present work.

2. Literature information

2.1. Ti–V system

The Ti–V system was optimized by Saunders [14] and Ghosh [15]. Phase equilibria of the Ti–V system were rather simple because there was no intermediate phase. It was optimized by Saunders [14] firstly. Ghosh [15] re-optimized this system and a good agreement was obtained between the assessment results and the experiment data. **The thermodynamic parameters assessed by Ghosh [15] were adopted in this work** and the calculated Ti–V phase diagram is shown in Fig. 1.

2.2. Ti–Zr system

The Ti–Zr system was optimized by Murry [16], Hari Kumar et al. [17] and Turchanin et al. [18]. During the work of Murry [16] and Hari Kumar et al. [17], the mixing enthalpy of the liquid alloys [19] in the Ti–Zr system was not considered, and then Turchanin et al. [18] calculated the binary system with the data of the mixing enthalpy of the liquid alloys [19]. Later on, the Ti–Zr system was re-optimized by Gao [20] during the process of thermodynamic assessment of the Sn–Ti–Zr system. Due to the narrow two-phase region (bcc+hcp) of the Ti–Zr binary system at 1023 and 1073 K calculated by the thermodynamic parameters of Gao [20], the bcc+hcp two-phase region in the Ti–V–Zr system could not be reproduced well, and it is necessary to re-optimize the thermodynamic parameters of hcp phase of Ti–Zr system. The calculated Ti–Zr phase diagram using the present thermodynamic parameters is shown in Fig. 2.

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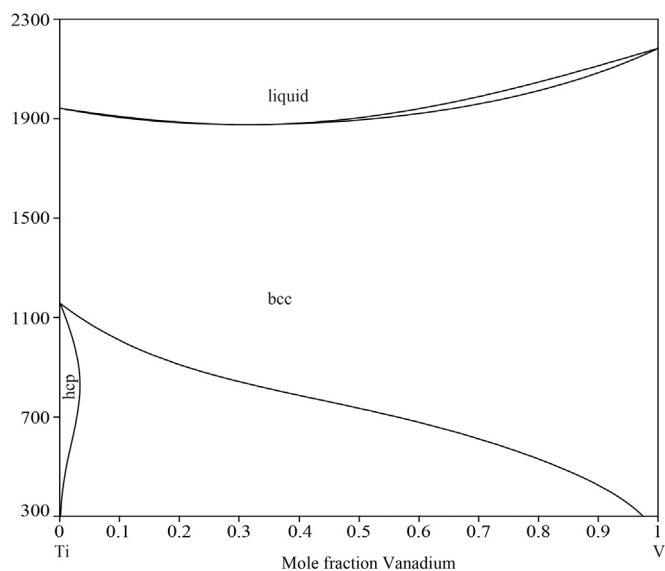


Fig. 1. Calculated Ti-V phase diagram using the thermodynamic parameters obtained by Ghosh [15].

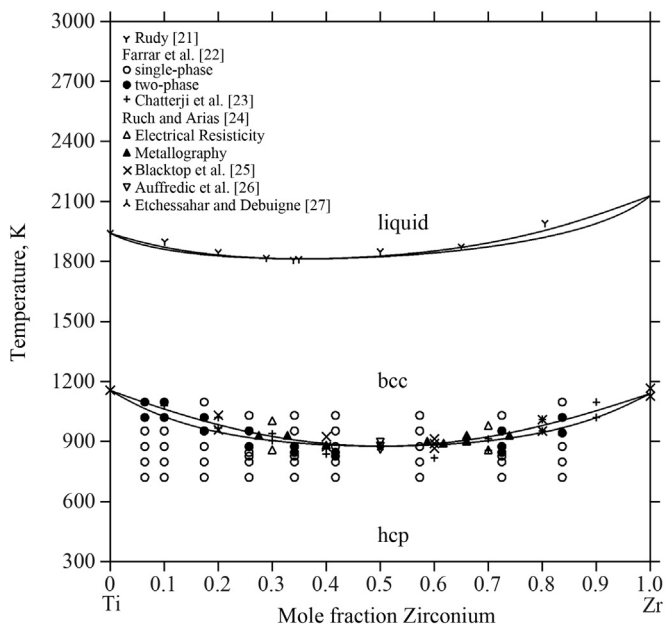


Fig. 2. Calculated Ti-Zr phase diagram using the modified thermodynamic parameters of hcp in the present work and thermodynamic parameters of bcc and liquid in Ref. [20] in comparison with the experimental data [21–27].

2.3. V-Zr system

The phase diagram of the V-Zr system was investigated experimentally by many researchers [28–31] and critically reviewed by Massalski and Okamoto [32]. This binary system was modeled firstly by Korb et al. [33]. Later, the V-Zr system was optimized by Servant [34] and Zhao et al. [35]. Based on the revised experimental data, Cui et al. [31] re-optimized the V-Zr system. In the present work, the V-Zr system was re-assessed on the basis of the thermodynamic description [31], because the phase relations between bcc and hcp phases in the Zr-rich corner of Ti-V-Zr system were difficult to be reproduced by means of thermodynamic parameters in Ref. [31]. The calculated V-Zr phase diagram using the present thermodynamic description in comparison with the experimental data is shown in Fig. 3.

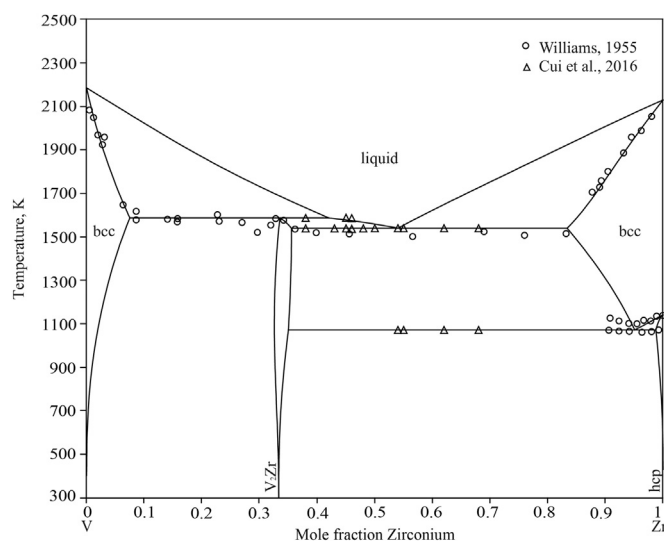


Fig. 3. Calculated V-Zr phase diagram using the present thermodynamic description in comparison with the experimental data [29,31].

2.4. Ti-V-Zr system

Nowikon and Baer [12] studied the Ti-V-Zr phase diagram and determined the isothermal section at 1023 K by metallography and XRD. Brunsch and Steeb [13] inferred the isothermal section at 1073 K from the data on the structure of diffusion zone during the investigation of diffusion behavior in this system by an electron microprobe. Based on the experimental data of Nowikon and Baer [12] and Brunsch and Steeb [13], Enomoto [36] critically reviewed the Ti-V-Zr ternary system, including the three constituent binary systems and isothermal sections of the Ti-V-Zr system.

3. Experimental procedures

Twenty six alloy compositions were selected to investigate the liquidus surface projection of the Ti-V-Zr system by XRD, SEM with EDS and DTA, in which 10 different compositions samples were used to study the isothermal sections of the Ti-V-Zr system at 1073 and 1273 K by XRD and SEM with EDS. The compositions of 26 alloys for the liquidus surface projection, and the chosen 10 samples for the isothermal sections of the Ti-V-Zr system at 1073 and 1273 K were detailed in Tables 1–3, respectively. The purity of Titanium, Vanadium and Zirconium was 99.995, 99.95 and 99.96 wt%, respectively. All samples were prepared by arc melting on a water-cooled copper hearth under the purity of argon atmosphere as 99.999%. Pure Ti was used as an oxygen getter, which was put into the crucible before all samples were melted, owing to the fact that pure Ti was prone to oxidation. To ensure homogeneity, the ingots were melted no less than four times and inverted after every melting. Samples with melting losses not exceeding 0.5 wt% were chosen for the experiments.

For the investigation of liquidus surface projection of the Ti-V-Zr system, the samples were cut into two halves using wire-electrode. One half was used for determining crystal structure by XRD [14] and microstructure by SEM. The XRD was used to identify the crystal structure of the constituent phases. The XRD measurement was carried out on a Rigaku rotatory anode harrow X-pert diffractometer using Cu K α radiation at 40 kV and 150 mA. The data was collected in the range of 2 θ from 20° to 100° with a step size of 0.02°. Meanwhile, the microstructure of the alloys was studied by means of SEM operating at 20 kV using a ZEISS EVO 18 Special Edition microscope. The other half of samples was cut into

Table 1

Alloy compositions of ternary Ti–V–Zr alloys selected for determining the liquidus surface.

No.	Alloy composition (at%)			Primary solidification phase	No.	Alloy composition (at%)			Primary solidification phase
	Ti	V	Zr			Ti	V	Zr	
#1	11.85	32.90	55.25	bcc(Zr)	#14	20.13	46.56	33.31	bcc(V)
#2	33.00	35.04	31.96	bcc(Zr)	#15	34.67	47.74	17.59	bcc(V)
#3	22.98	37.23	39.79	bcc(Zr)	#16	19.80	50.30	29.90	bcc(V)
#4	31.96	40.02	28.02	bcc(Zr)	#17	15.78	52.38	31.84	bcc(V)
#5	10.93	40.20	48.87	bcc(Zr)	#18	18.85	53.33	27.82	bcc(V)
#6	5.82	39.19	54.99	bcc(Zr)	#19	22.75	53.21	24.04	bcc(V)
#7	24.65	43.92	31.43	bcc(Zr)	#20	11.81	57.24	30.95	bcc(V)
#8	18.85	43.17	37.98	bcc(Zr)	#21	6.77	56.29	36.94	bcc(V)
#9	6.16	43.59	50.25	bcc(Zr)	#22	4.84	57.36	37.80	bcc(V)
#10	1.91	43.17	54.92	bcc(Zr)	#23	10.80	59.26	29.94	bcc(V)
#11	6.03	46.73	47.24	V ₂ Zr	#24	11.70	63.30	25.00	bcc(V)
#12	3.08	47.74	49.18	V ₂ Zr	#25	9.73	65.32	24.95	bcc(V)
#13	3.00	49.82	47.18	V ₂ Zr	#26	40.85	48.09	11.06	bcc(V)

Table 2

Summary of identified phases and their compositions for the Ti–V–Zr alloys annealed at 1073 K for 30 days.

No.	Alloy composition (at%)			Phases	Phase composition (at%)		
	Ti	V	Zr		Ti	V	Zr
#1	11.85	32.90	55.25	bcc(Zr)	16.21	6.28	77.51
				V ₂ Zr	5.86	61.22	32.92
#3	22.98	37.23	39.79	bcc(Zr)	31.43	8.49	60.08
				bcc(V)	14.02	84.17	1.81
				V ₂ Zr	9.27	58.52	32.21
#4	31.96	40.02	28.02	bcc(V)	25.47	73.28	1.25
				bcc(Zr)	38.27	10.15	51.58
#5	10.93	40.20	48.87	bcc(Zr)	19.47	6.70	73.83
				V ₂ Zr	6.65	61.22	32.13
#6	5.82	39.19	54.99	bcc(Zr)	14.2	12.57	73.23
				V ₂ Zr	6.01	62.51	31.48
#13	3.00	49.82	47.18	bcc(Zr)	6.37	5.63	88.00
				V ₂ Zr	4.62	65.30	30.08
				hcp(Zr)	0.93	2.5	96.57
#17	15.78	52.38	31.84	bcc(Zr)	27.82	15.18	57
				bcc(V)	13.37	85.85	0.78
				V ₂ Zr	8.64	61.2	30.16
#19	22.75	53.21	24.04	bcc(V)	15.09	84.55	0.36
				bcc(Zr)	28.88	15.35	55.77
#25	9.73	65.32	24.95	bcc(Zr)	31.21	8.53	60.26
				bcc(V)	12.22	86.33	1.45
				V ₂ Zr	7.69	60.21	32.10
#26	40.85	48.09	11.06	bcc(V)	40.16	57.79	2.05
				bcc(Ti)	50.18	13.77	36.05

flakelets for determining phase transition temperature by DTA, which was performed on a SDT Q600 (TA instruments). Zirconia crucibles were used and measurements were performed under flowing argon atmosphere. In order to obtain the better thermal conduction and thermal diffusion, the samples were cut into slices that just covered the bottom of the crucible and the heating rate was as slow as 10°C/min. A Pt–Pt/Rh thermocouple was used to monitor temperature. In the examined temperature range, the accuracy of the temperature measurement was estimated to be $\pm 0.5^\circ$.

The chosen samples for determining isothermal sections at 1073 and 1273 K were sealed in quartz tubes pre-evacuated and refilled with high-purity argon, and then the samples were annealed at 1073 K for 30 days and 1273 K for 15 days in a box – type furnace (KSL–1200X–A1 and KSL–1400X–A1). After annealing, the ampoules with samples were quenched in ice water. Each quenched sample for the ternary system was investigated by XRD and SEM with EDS to determine the crystal structure and compositions of the constituent phases.

Table 3

Summary of identified phases and their compositions for the Ti–V–Zr alloys annealed at 1273 K for 15 days.

No.	Alloy composition (at%)			Phases	Phase composition (at%)		
	Ti	V	Zr		Ti	V	Zr
#1	11.85	32.90	55.25	bcc(Zr)	15.19	10.84	73.97
				V ₂ Zr	5.78	63.71	30.51
#3	22.98	37.23	39.79	bcc(Zr)	27.62	14.71	57.67
				bcc(V)	15.18	83.92	0.9
				V ₂ Zr	9.69	60	30.31
#4	31.96	40.02	28.02	bcc(V)	24.58	74.71	0.71
				bcc(Zr)	33.26	16.63	50.11
#5	10.93	40.20	48.87	bcc(Zr)	7.72	11.28	81
				V ₂ Zr	3.52	63.85	32.63
#6	5.82	39.19	54.99	bcc(Zr)	14.2	12.57	73.23
				V ₂ Zr	6.01	62.51	31.48
#13	3.00	49.82	47.18	bcc(Zr)	5.7	3.28	91.02
				V ₂ Zr	3.52	63.85	32.63
#17	15.78	52.38	31.84	bcc(Zr)	27.82	15.18	57
				bcc(V)	13.37	85.85	0.78
				V ₂ Zr	8.64	61.2	30.16
#19	22.75	53.21	24.04	bcc(V)	15.09	84.55	0.36
				bcc(Zr)	28.88	15.35	55.77
#25	9.73	65.32	24.95	bcc(Zr)	27.09	15.07	57.84
				bcc(V)	13.19	86.03	0.78
				V ₂ Zr	6.98	61.92	31.1
#26	40.85	48.09	11.06	bcc(V)	39.36	55.83	4.81
				bcc(Ti)	45.4	25.19	29.41

4. Thermodynamic modeling

4.1. Thermodynamic model

In the present work, the Gibbs energy functions for the unary phases of elements Ti, V and Zr are taken from the SGTE (Scientific Group Thermodata Europe) database of pure elements [37].

In the Ti–V–Zr system, there are three solution phases, liquid, bcc and hcp, which are modeled with substitutional solution models as (Ti,V,Zr), (Ti,V,Zr)₁(Va)₃ and (Ti,V,Zr)₁(Va)_{0.5}.

The compound V₂Zr in the V–Zr system has a homogeneity range and is treated as (Ti,V,Zr)₂(Ti,V,Zr)₁ by a two-sublattice model [38,39].

4.2. Modeling procedure

The optimization is carried out by means of the optimization module PARROT [40] of the thermodynamic software ThermoCalc, which works by minimizing the square sum of the differences between experimental and computed values. In the

assessment process, each piece of experimental information is given a certain weight according to its estimated accuracy. Owing to the integrity and accuracy of the experimental data obtained in present work, the present experimental data is given a larger weight. The weights were changed systematically during the assessment until most of the experimental data were accounted for within the expected uncertainty limits.

Based on the experimental isothermal sections at 1273 and 1373 K from this work and at 1023 K from Nowikon and Baer [12], the ternary interaction parameters of solid solution phases, bcc and hcp, and V_2Zr phase are assessed on the phase relations among each phase. During the process of optimization, the most difficult work is to reproduce the miscibility gap of bcc phase. Next, the liquidus surface projection and invariant equilibria are taken into account on the basis of the experimental data from the present work. In this process, three thermodynamic parameters for the liquid, ${}^0L_{Ti,V,Zr}^{liq}$, ${}^1L_{Ti,V,Zr}^{liq}$ and ${}^2L_{Ti,V,Zr}^{liq}$, are assessed to reproduce the experimental data. Finally, all of thermodynamic parameters are slightly changed to make the phase relations and liquidus surface consistent with experimental data.

5. Results and Discussion

5.1. Experimental investigation

Detailed experimental results from XRD, SEM-EDS and DTA are presented in Tables 1–4.

5.1.1. Liquidus surface projection of the Ti–V–Zr system

The primary solidification of the bcc(Zr) phase is observed for alloys #1–#10. Fig. 4a shows Backscattered Electron Image of SEM (BEI) micrograph of alloy #1 (V–32.90 at.% Ti–55.25 at.% Zr). Three different microstructure regions, the primary solidification phase, the binary eutectic microstructure and ternary eutectic microstructure, are observed. In the microstructure, the bright dendrite microstructure is the primary solidification phase, which composition is Ti–10.05 at.%V–80.1 at.%Zr by EDS, and is the bcc (Zr) phase. The solidification path of samples #1 would be following the sequence: liquid $\rightarrow$ bcc(Zr) $\rightarrow$ bcc(Zr) + bcc(V) $\rightarrow$ bcc (Zr) + bcc(V) + V_2Zr .

For the alloys #11–#13, the primary solidification phase identified is V_2Zr as shown in Fig. 4b and c. Fig. 4b shows the diffraction pattern of alloy #13 (Ti–49.82 at.%V–47.18 at.%Zr), and the diffraction peaks of bcc(V), V_2Zr and bcc(Zr) phases are observed. Fig. 4c is the BEI micrograph for alloy #13. Three different microstructure regions are observed. It can be identified from the microstructure that the thick dark microstructure is the primary solidification phase, which is the first phase precipitated from liquid phase during solidification. Its composition measured by EDS is Ti–65.06 at.%V–32.15 at.%Zr, and it is the V_2Zr phase. The coarse lamellar chequered with dark and bright regions is corresponding to the formation of the binary eutectic structure (V_2Zr + bcc(Zr)). The thinnest lamellar microstructure is the ternary eutectic structure bcc(V) + V_2Zr + bcc(Zr). The XRD result is consistent with the metallographic analysis. According to these analysis procedures, the solidification path of samples #13 would be following the sequence: liquid $\rightarrow$ V_2Zr $\rightarrow$ bcc(Zr) + V_2Zr $\rightarrow$ bcc

(Zr) + bcc(V) + V_2Zr .

Fig. 4d is the BEI micrograph for alloy #14 (Ti–46.56 at.% V–33.31 at.%Zr). Two kinds of typical eutectic microstructure are observed in sample #14. Based on the metallographic analysis, The coarse lamellar chequered with dark and bright regions is the microstructure of two phases bcc(Zr) + bcc(V) and the gray acicular region is the ternary eutectic microstructure (bcc(Zr) + bcc(V) + V_2Zr). According to its observed metallographic microstructure, the composition of alloy #14 is located near to the monovariant line (liquid $\rightarrow$ bcc(Zr) + bcc(V)).

The bcc(V) phase is the primary solidification phase in many samples (#15–#26). Fig. 4f showed microstructure in the BEI micrograph of alloy #21. Three different microstructure regions with typical peritectic characteristic, the dark microstructure, the gray microstructure and the eutectic microstructure are observed. The dark dendritic microstructure is the primary solidification phase, which composition is Ti–95.53 at.%V–0.95 at.%Zr by EDS, and is the bcc(V) phase. The gray microstructure around the dark dendritic microstructure is the secondary solidification phase, which is formed by means of peritectic reaction liquid + bcc(V) $\rightarrow$ V_2Zr . The composition of the gray microstructure is Ti–61.45 at.% V–35.69 at.%Zr by EDS, and it is the V_2Zr phase. The lamellar chequered with gray and bright regions is the ternary eutectic microstructure bcc(V) + bcc(Zr) + V_2Zr . According to its metallographic microstructure, the solidification path of the alloy #21 can be confirmed. The bcc (V) phase firstly precipitates from the liquid phase, then the V_2Zr phase is generated by the peritectic reaction liquid + bcc(V) $\rightarrow$ V_2Zr , and the solidification path of the alloy #21 is finished with the ternary eutectic reaction E_1 : liquid $\rightarrow$ bcc(V) + bcc (Zr) + V_2Zr as listed in Table 4. These results are in agreement with the XRD diffractogram of alloy #21 as shown in Fig. 4e, in which the diffraction peaks of bcc(V), bcc(Zr) and V_2Zr are observed.

The temperature of the invariant reactions and the direction of the temperature descent for the monovariant lines are determined using thermal analysis results. Fig. 5 gives the DTA heating curves of alloy #5 (Ti–40.20 at.%V–48.87 at.%Zr) and alloy #18 (Ti–53.33 at.%V–27.82 at.%Zr) with a heating rate of 10 K/min. Both of the curves show two clearly visible peaks. According to the BEI analysis on the alloys and the phase equilibria of the Ti–V–Zr system, the peaks at 805 and 803 K are corresponding to the temperature of the invariant reaction V_2Zr + bcc(Zr) $\rightarrow$ bcc(V) + hcp (Zr) and the peak at 1485 K corresponds to the temperature of invariant reaction liquid $\rightarrow$ bcc(V) + bcc(Zr) + V_2Zr .

5.1.2. Isothermal section of the Ti–V–Zr system at 1073 K

The chosen 10 samples of the Ti–V–Zr ternary system are prepared and annealed at 1073 K for 30 days. The compositions of the samples are summarized in Table 2. V-rich solid solution (bcc (V)), Zr-rich solid solution (bcc (Zr) and hcp(Zr)), Ti-rich solid solution (bcc (Ti)) and V_2Zr are observed in the isothermal section of the Ti–V–Zr system at 1073 K.

Fig. 6a and b shows the X-ray diffractogram and the BEI micrograph of alloys #5 (Ti–40.20 at.% V–48.87 at.% Zr). Two different phases are found in alloys #5. Based on the XRD results and the BEI micrograph of alloy #5 as shown in Fig. 6a and b, alloy #5 consists of two different phases V_2Zr and bcc(Zr) as listed in Table 3. The use of similar analysis procedures shows that alloys #1 and #6 are all located within the bcc(Zr) + V_2Zr two-phase region.

Fig. 6c shows the X-ray diffractogram of alloy #13 (Ti–49.82 at.% V–47.18 at.% Zr). XRD analysis indicates that it contains three phases, hcp (Zr), bcc (Zr), and V_2Zr . The microstructure of this alloy is shown in Fig. 6d. Three different microstructure regions can be observed. The compositions of the dark region, the gray region and the bright region are Ti–65.30 at.% V–30.08 at.% Zr, Ti–2.5 at.% V–96.57 at.% Zr, and Ti–5.63 at.% V–88.00 at.% Zr by EDS, which are corresponding to V_2Zr , hcp(Zr) and bcc(Zr), respectively. The

Table 4
Invariant reactions in the Ti–V–Zr system.

Type	Invariant reaction	Temperature(K)	DTA results
E_1	liquid $\rightarrow$ bcc(V) + bcc(Zr) + V_2Zr	1485	1485
U_1	V_2Zr + bcc(Zr) $\rightarrow$ bcc(V) + hcp(Zr)	804	804

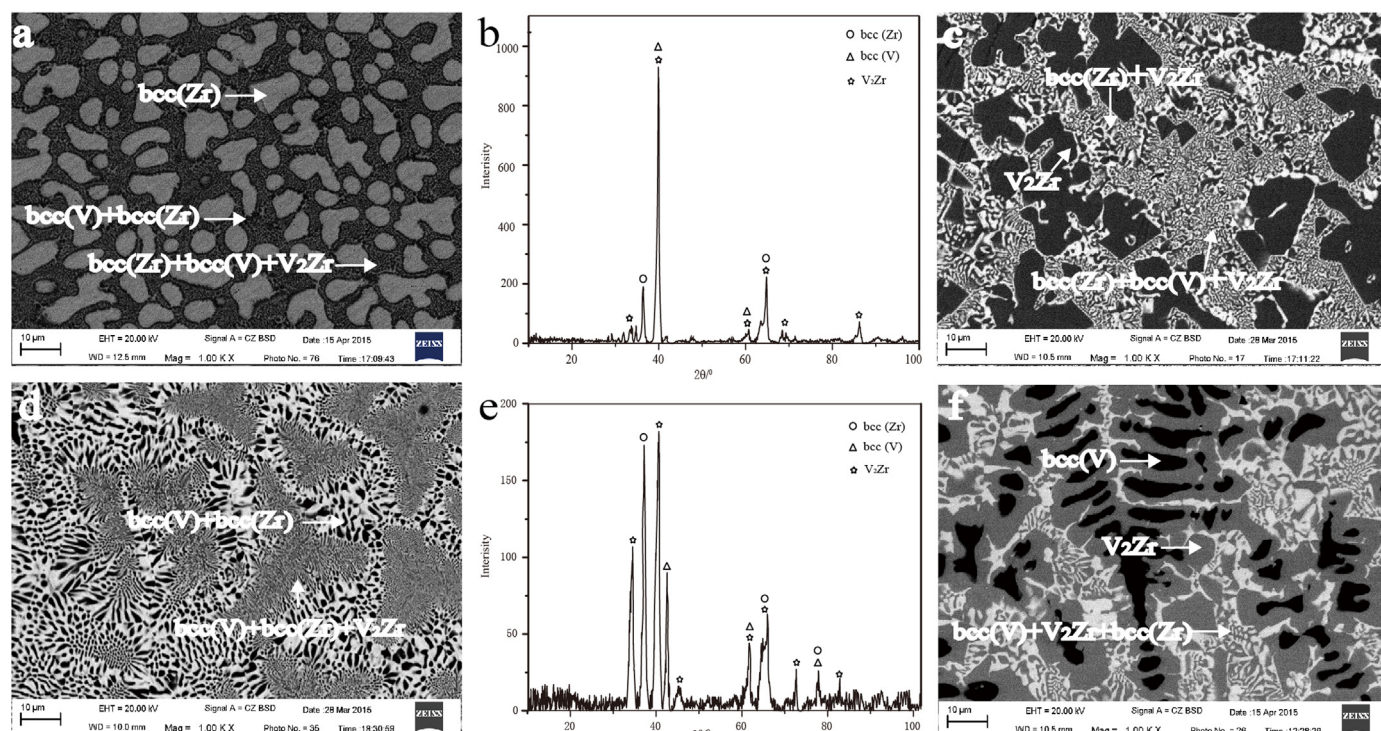


Fig. 4. (a) BEI micrograph of alloy #1 (Ti–32.90 at.%V–55.25 at.%Zr) (b) XRD diffraction pattern of alloy #13 (Ti–49.82 at.%V–47.18 at.%Zr) (c) BEI micrograph of alloy #13 (Ti–49.82 at.%V–47.18 at.%Zr) (d) BEI micrograph of alloy #14 (Ti–46.56 at.%V–33.31 at.%Zr) (e) XRD diffraction pattern of alloy #21 (Ti–56.29 at.%V–36.94 at.%Zr) (f) BEI micrograph of alloy #21 (Ti–56.29 at.%V–36.94 at.%Zr).

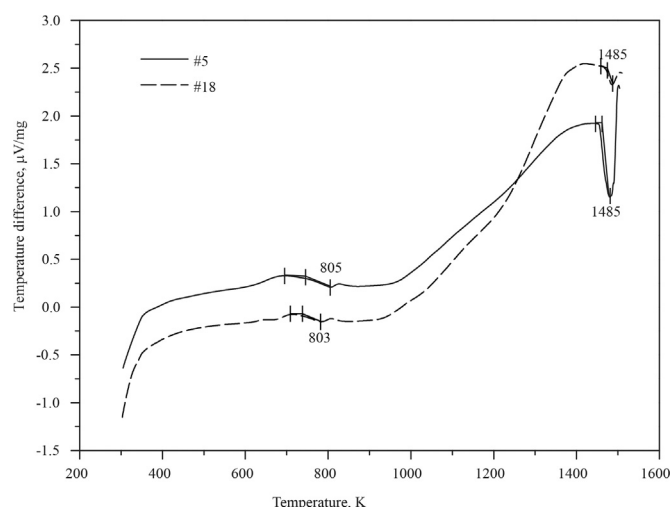


Fig. 5. DTA heating curves of alloy #5 (Ti–40.20 at.%V–48.87 at.%Zr) and alloy #18 (Ti–53.33 at.%V–27.82 at.%Zr).

XRD results are in agreement with the microstructure analysis. So alloy #13 is located in the $V_2Zr + bcc(Zr) + hcp(Zr)$ three-phases region.

As shown in Fig. 6e and f, the peaks of bcc (V), bcc (Zr), and V_2Zr are found in the XRD diffraction pattern of alloy #25. There are three different microstructure regions in this alloy BEI micrograph. The compositions of the identified phase are listed in Table 3. According to the results of XRD diffraction pattern and BEI micrograph, alloy #25 is in the $bcc(Zr) + bcc(V) + V_2Zr$ three-phase region. The similar results are confirmed in alloys #3 and #17.

Fig. 6g shows the XRD diffractogram for alloy #26 (Ti–48.09 at.% V–11.06 at.% Zr), the diffraction peaks of bcc(V) and bcc(Ti) are

observed. Fig. 6h shows the BEI micrograph for alloy #26. The compositions of the dark phase and the bright phase are Ti–57.79 at.% V–2.05 at.% Zr and Ti–13.77 at.% V–36.05 at.% Zr by EDS, respectively. Combined with the results of XRD diffractogram, it is thus presumed that the dark region is bcc (V) phase and the bright region is bcc(Ti) phase. It is within the $bcc(V) + bcc(Ti)$ two-phases field. Similar results are found in alloys #4 and #19.

No ternary compounds are observed at 1073 K in the Ti–V–Zr system.

5.1.3. Isothermal section of the Ti–V–Zr system at 1273 K

The chosen 10 samples of the Ti–V–Zr system are prepared and annealed at 1273 K for 15 days.

Fig. 7a and b shows the X-ray diffractogram and the BEI micrograph of alloys #1 (Ti–32.90 at.% V–55.25 at.% Zr). Two phases (V_2Zr and $bcc(Zr)$) are found in alloy #1, and the phase compositions are listed in Table 3. The use of similar analysis procedures shows that alloys #5, #6 and #13 are all located within the $bcc(Zr) + V_2Zr$ two-phases region.

Fig. 7c shows the X-ray diffractogram of alloy #25 (Ti–65.32 at.% V–24.95 at.% Zr). XRD analysis indicates that it contains three phases, bcc (V), V_2Zr , and bcc (Zr). The microstructure of this alloy is shown in Fig. 7d. Three different microstructure regions can be observed. The compositions of the dark region, the gray region and the bright region are Ti–86.03 at.% V–0.78 at.% Zr, Ti–61.92 at.% V–31.1 at.% Zr, and Ti–15.07 at.% V–57.84 at.% Zr by EDS, which are bcc(V), V_2Zr , and bcc(Zr), respectively. The XRD results are in agreement with the microstructure analysis. Similar results are observed for alloys #3 and #17, which are all within the $bcc(Zr) + bcc(V) + V_2Zr$ three-phases region at 1273 K.

Fig. 7e–f shows the XRD diffractogram and BEI micrograph for alloy #26 (Ti–48.09 at.% V–11.06 at.% Zr). It is within the $bcc(V) + bcc(Ti)$ two-phases field. Similar results are found in alloys #4 and #19.

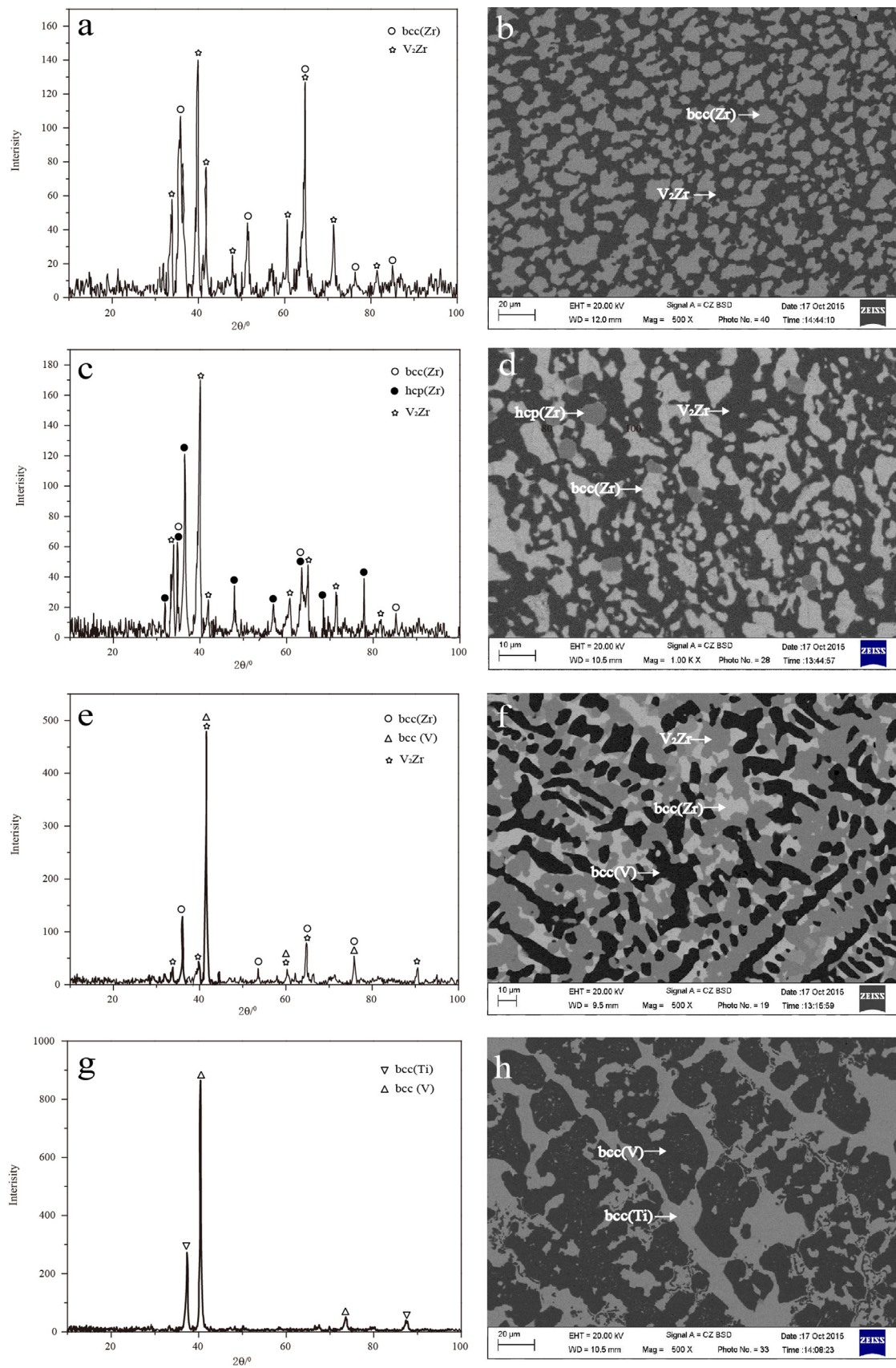


Fig. 6. Partial XRD diffraction pattern and BEI micrograph of alloys annealed at 1073 K for 30 days. (a) XRD diffraction pattern of alloy #5 (Ti–40.20 at.% V–48.87 at.% Zr) (b) BEI micrograph of alloy #5 (Ti–40.20 at.% V–48.87 at.% Zr) (c) XRD diffraction pattern of alloy #13 (Ti–49.82 at.% V–47.18 at.% Zr) (d) BEI micrograph of alloy #13 (Ti–49.82 at.% V–47.18 at.% Zr) (e) XRD diffraction pattern of alloy #25 (Ti–65.32 at.% V–24.95 at.% Zr) (f) BEI micrograph of alloy #25 (Ti–65.32 at.% V–24.95 at.% Zr) (g) XRD diffraction pattern of alloy #26 (Ti–48.09 at.% V–11.06 at.% Zr) (h) BEI micrograph of alloy #26 (Ti–48.09 at.% V–11. at.% Zr).

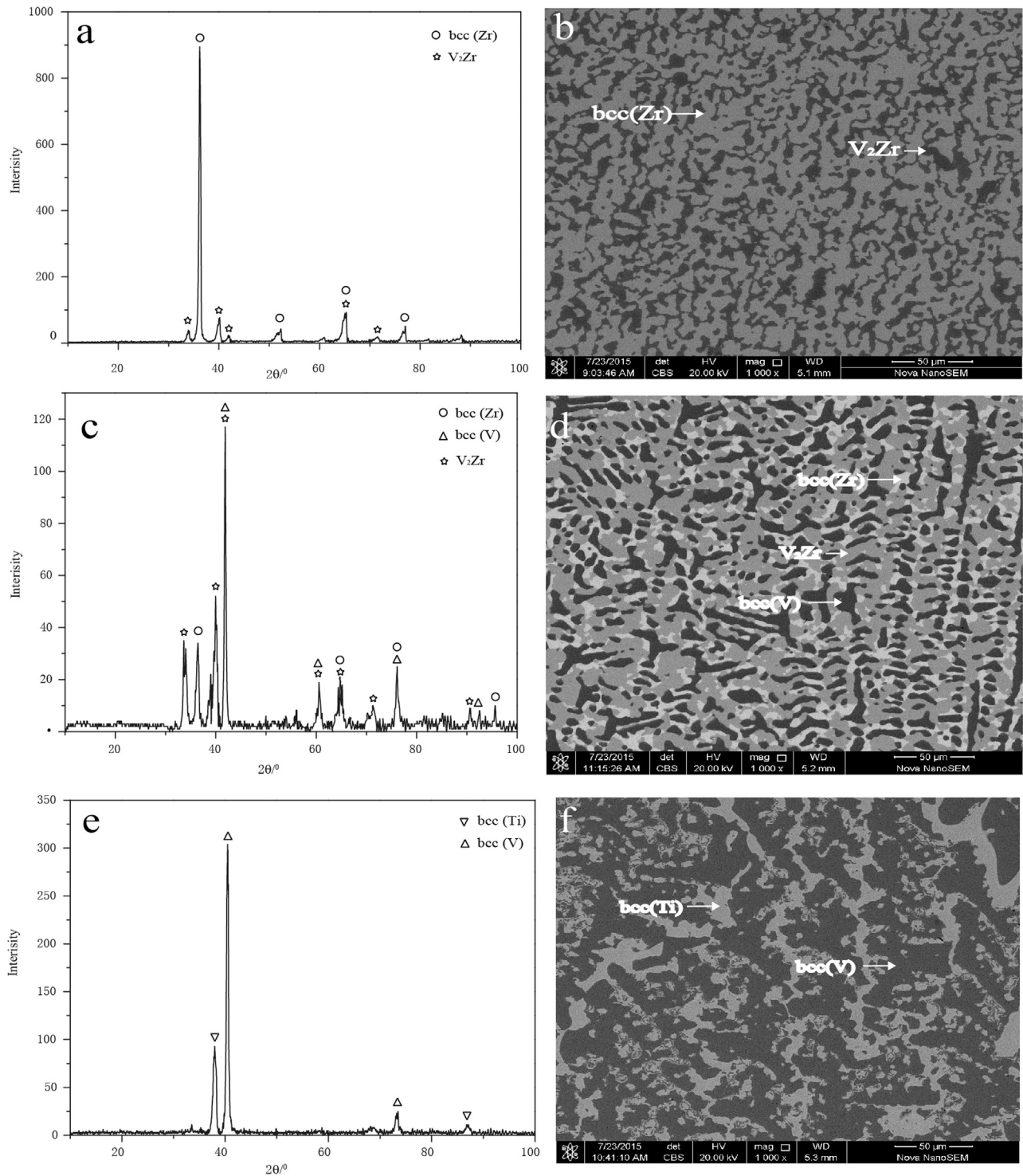


Fig. 7. Partial XRD diffraction pattern and BEI micrograph of alloys annealed at 1273 K for 15 days. (a) XRD diffraction pattern of alloy #1 (Ti–32.90 at.% V–55.25 at.% Zr) (b) BEI micrograph of alloy #1 (Ti–32.90 at.% V–55.25 at.% Zr) (c) XRD diffraction pattern of alloy #25 (Ti–65.32 at.% V–24.95 at.% Zr) (d) BEI micrograph of alloy #25 (Ti–65.32 at.% V–24.95 at.% Zr) (e) XRD diffraction pattern of alloy #26 (Ti–48.09 at.% V–11.06 at.% Zr) (f) BEI micrograph of alloy #26 (Ti–48.09 at.% V–11.06 at.% Zr).

No ternary compounds are also observed at 1273 K in the Ti–V–Zr system.

5.2. Thermodynamic optimization

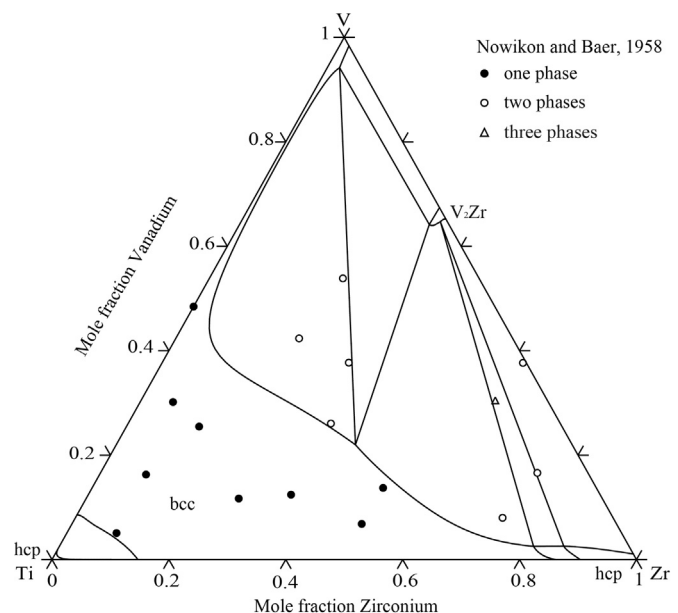
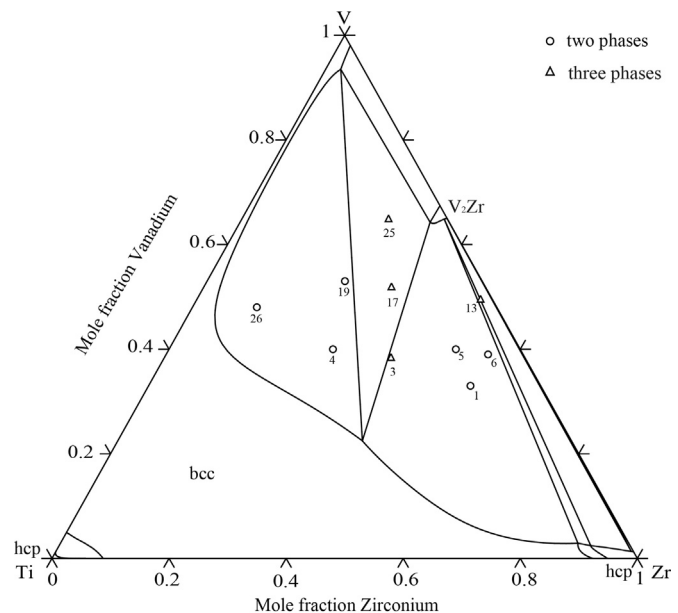
A thermodynamic optimization the Ti–V–Zr ternary system is carried out using the CALPHAD method. The invariant reactions determined in the presented work are listed in Table 4.

As shown in Table 4, the satisfactory agreement is presented between the calculated invariant reaction temperature and the present experimental results. The thermodynamic parameters of the Ti–V–Zr system obtained in the present work are listed in Table 5.

Figs. 8–10 represent the calculated isothermal sections of the Ti–V–Zr system at 1023, 1073 and 1273 K in comparison with experimental data. Obviously, the calculated result is in good

Table 5
Thermodynamic parameters in the Ti–V–Zr system.^a

Phase	Thermodynamic parameters	Refs.
liquid	model (Ti, V, Zr) ₁	
	$0_{\text{Ti,V}}^{\text{liq}} = 368.55$	[15]
	$1_{\text{Ti,V}}^{\text{liq}} = 2838.63$	[15]
	$0_{\text{V,Zr}}^{\text{liq}} = -15999.99 + 13.6515T$	This work
	$1_{\text{V,Zr}}^{\text{liq}} = 2835.36 - 1.9742T$	This work
	$2_{\text{V,Zr}}^{\text{liq}} = -4176.74$	This work
	$0_{\text{Ti,Zr}}^{\text{liq}} = -21888.76 - 1.0100T$	[20]
	$0_{\text{Ti,V,Zr}}^{\text{liq}} = -174029.31 - 8.7276T$	This work
	$1_{\text{Ti,V,Zr}}^{\text{liq}} = -34764.05 + 4.0897T$	This work
	$2_{\text{Ti,V,Zr}}^{\text{liq}} = 47326.08$	This work
bcc (V)	model (Ti, V, Zr) ₁ (Va) ₃	
	$0_{\text{Ti,V}}^{\text{bcc}} = 6523.17$	[15]
	$1_{\text{Ti,V}}^{\text{bcc}} = 2025.39$	[15]
	$0_{\text{V,Zr}}^{\text{bcc}} = 17872.99 + 8.7539T$	This work
	$1_{\text{V,Zr}}^{\text{bcc}} = -3208.60 + 4.8481T$	This work
	$0_{\text{Ti,Zr}}^{\text{bcc}} = -48287.53 + 17.4484T$	[20]
	$0_{\text{Ti,V,Zr}}^{\text{bcc}} = -253783.69 + 10.9738T$	This work
	$1_{\text{Ti,V,Zr}}^{\text{bcc}} = 7589.27$	This work
	$2_{\text{Ti,V,Zr}}^{\text{bcc}} = 80496.47$	This work
hcp (Ti and Zr)	model (Ti, V, Zr) ₁ (Va) _{0.5}	
	$0_{\text{Ti,V}}^{\text{hcp}} = 13233$	[15]
	$0_{\text{V,Zr}}^{\text{hcp}} = 116615.15 + 2.8664T$	This work
	$1_{\text{V,Zr}}^{\text{hcp}} = 89336.23 + 5.0607T$	This work
	$0_{\text{Ti,Zr}}^{\text{hcp}} = -81673.00 + 60.7792T$	This work
	$2_{\text{Ti,Zr}}^{\text{hcp}} = 977.17 - 11.5750T$	This work
	$0_{\text{Ti,V,Zr}}^{\text{hcp}} = 581627.28$	This work
	$2_{\text{Ti,V,Zr}}^{\text{hcp}} = -348585.54$	This work
V ₂ Zr	model (Ti, V, Zr) ₂ (Ti, V, Zr)	
	$G_{\text{Ti:Ti}}^{\text{V}_2\text{Zr}} = 3\text{GHSE}_{\text{Ti}} + 15000.00$	This work
	$G_{\text{Ti:V}}^{\text{V}_2\text{Zr}} = 2\text{GHSE}_{\text{Ti}} + \text{GHSE}_{\text{V}} + 30000.00$	This work
	$G_{\text{Ti:Zr}}^{\text{V}_2\text{Zr}} = 2\text{GHSE}_{\text{Ti}} + \text{GHSE}_{\text{Zr}} + 15000.00$	This work
	$G_{\text{V:Ti}}^{\text{V}_2\text{Zr}} = 2\text{GHSE}_{\text{V}} + \text{GHSE}_{\text{Ti}} + 30000.00$	This work
	$G_{\text{V:V}}^{\text{V}_2\text{Zr}} = 3\text{GHSE}_{\text{V}} + 15000.00$	[31]
	$G_{\text{V:Zr}}^{\text{V}_2\text{Zr}} = 2\text{GHSE}_{\text{V}} + \text{GHSE}_{\text{Zr}} - 10796.71 + 3.8144T$	This work
	$G_{\text{Zr:Ti}}^{\text{V}_2\text{Zr}} = 2\text{GHSE}_{\text{Zr}} + \text{GHSE}_{\text{Ti}} + 15000.00$	This work
	$G_{\text{Zr:V}}^{\text{V}_2\text{Zr}} = 2\text{GHSE}_{\text{Zr}} + \text{GHSE}_{\text{V}} + 40796.71 - 3.8144T$	This work
	$G_{\text{Zr:Zr}}^{\text{V}_2\text{Zr}} = 3\text{GHSE}_{\text{Zr}} + 15000.00$	[31]
	$0_{\text{Ti,V,Zr}}^{\text{V}_2\text{Zr}} = 0_{\text{V,V,Zr}}^{\text{V}_2\text{Zr}} = 0_{\text{Zr,V,Zr}}^{\text{V}_2\text{Zr}} = 4086.75 + 9.5308T$	This work
	$0_{\text{V,Zr,Ti}}^{\text{V}_2\text{Zr}} = 0_{\text{V,Zr,V}}^{\text{V}_2\text{Zr}} = 0_{\text{V,Zr,Zr}}^{\text{V}_2\text{Zr}} = 35860.62 + 12.7268T$	This work
	$0_{\text{Ti,V,Ti}}^{\text{V}_2\text{Zr}} = 0_{\text{Ti,V,V}}^{\text{V}_2\text{Zr}} = 0_{\text{Ti,V,Zr}}^{\text{V}_2\text{Zr}} = -36973.73 + 15.1442T$	This work

^a In J mol⁻¹ of the formula units.**Fig. 8.** Calculated isothermal section of the Ti–V–Zr system at 1023 K using the present thermodynamic description in comparison with the experimental data from Nowikow and Baer [12].**Fig. 9.** Calculated isothermal section of the Ti–V–Zr system at 1073 K using the present thermodynamic description in comparison with the present experimental results.

and experimental data. The complete reaction scheme in the Ti–V–Zr system is shown in Fig. 12.

6. Conclusion

The liquidus surface projection of the Ti–V–Zr system was experimentally determined by XRD, SEM-EDS and DTA. There were three primary solidification regions, bcc (V), bcc (Zr) and V₂Zr. The temperatures of two invariant reactions, liquid → bcc(V) + bcc(Zr) + V₂Zr at 1485 K and V₂Zr + bcc(Zr) → bcc(V) + hcp(Zr) at 803 K, were confirmed. Two isothermal sections of the Ti–V–Zr system at 1073 and 1273 K were constructed according to the experimental phase

agreement with the corresponding experiment data in this work. Owing to the existence of bcc(Zr) phase in the V–Zr binary system at 1073 K, one more hcp(Zr) + bcc(Zr) + V₂Zr three-phase region is shown in the isothermal section of the Ti–V–Zr system at 1073 K.

Fig. 11 is the calculated liquidus projection of the Ti–V–Zr system using the present thermodynamic description in comparison with the present experimental results. The result shows that a satisfactory agreement is obtained between the calculated result

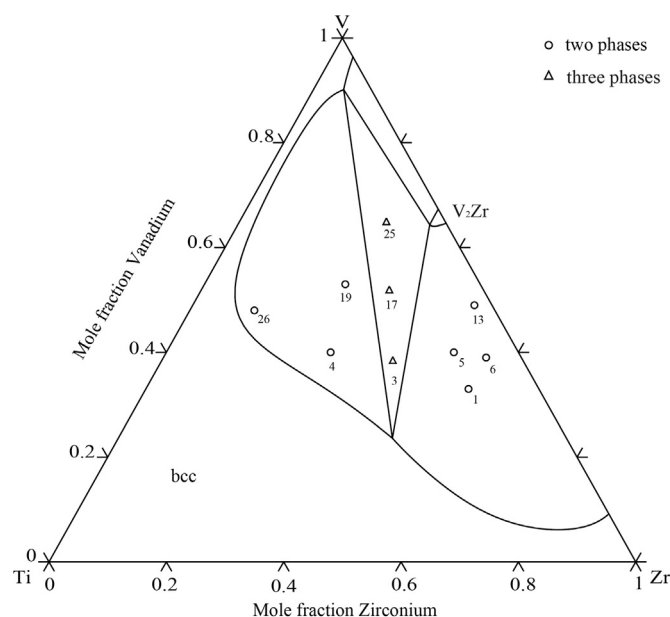


Fig. 10. Calculated isothermal section of the Ti-V-Zr system at 1273 K using the present thermodynamic description in comparison with the present experimental results.

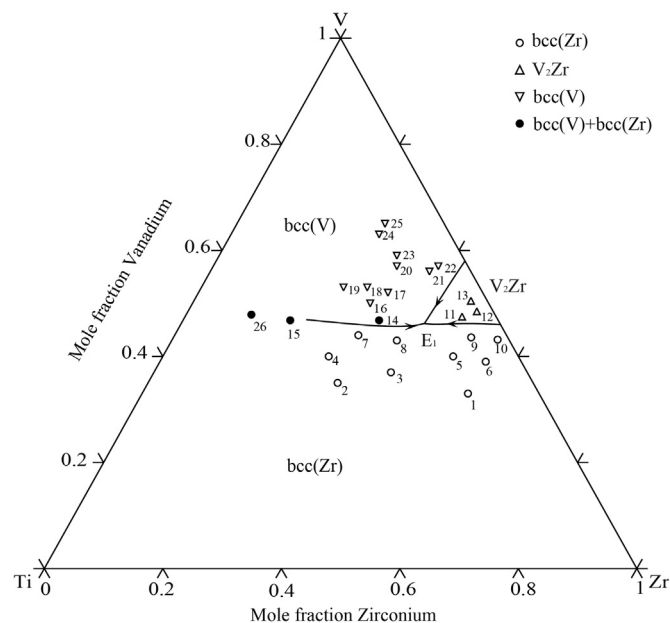


Fig. 11. Calculated liquidus projection of the Ti-V-Zr system using the present thermodynamic description in comparison with the present experimental results.

equilibria obtained by XRD and SEM-EDS. One three-phases region ($\text{bcc}(\text{Zr}) + \text{bcc}(\text{V}) + \text{V}_2\text{Zr}$) and two two-phases regions ($\text{bcc}(\text{Zr}) + \text{bcc}(\text{V})$ and $\text{bcc}(\text{Zr}) + \text{V}_2\text{Zr}$) were determined experimentally at 1273 K. Two three-phases regions $\text{bcc}(\text{Zr}) + \text{bcc}(\text{V}) + \text{V}_2\text{Zr}$ and $\text{bcc}(\text{Zr}) + \text{hcp}(\text{Zr}) + \text{V}_2\text{Zr}$ and two two-phases regions ($\text{bcc}(\text{Zr}) + \text{bcc}(\text{V})$ and $\text{bcc}(\text{Zr}) + \text{V}_2\text{Zr}$) were determined at 1073 K. No ternary compounds were found in experimental isothermal sections. Based on the experimental data in present work and literature, the thermodynamic optimization of the Ti-V-Zr system was carried out by the CALPHAD method. A set of self-consistent thermodynamic description for the Ti-V-Zr ternary system was obtained.

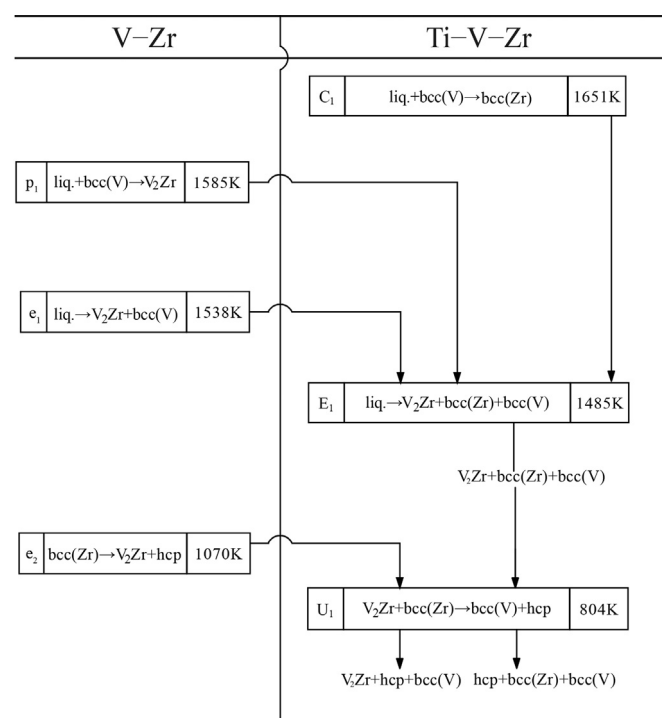


Fig. 12. Predicted reaction scheme in the Ti-V-Zr system.

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